State estimation from sparse sensors

Scientific question: can noisy point measurements correct a reduced cylinder-wake forecast without seeing future observations? We separate four information contracts: autonomous open-loop dynamics, causal filtering, instantaneous sensor-only reconstruction, and offline smoothing. Only the first three are evaluated here; smoothing is defined but not used.

Data are prior FlowMLLab Re110 LBM transverse-velocity fields on a 32 x 78 wake ROI. They are finite-resolution educational CFD, not exact continuum truth. The cylinder wall is outside the ROI. Synthetic sensor noise does not create independent CFD realizations.

Prerequisites: POD, least squares, Gaussian conditioning and Week 7 modal forecasting. CPU runtime is about 15 seconds.

Definitions and notation. POD (proper orthogonal decomposition): the singular value decomposition of the matrix of training snapshots; its spatial singular vectors Phi(n,r) are the modes. The column projection x_k = Phi^T (q_k - mean) gives the reduced state; the code stores the row a_row,k = x_k^T. ROI: the region of interest, the 32 x 78 wake window on which all fields are stored. DMD (dynamic mode decomposition): a linear model of the coefficient dynamics, a_row,k+1 = a_row,k A, fitted by least squares on training frames; "open-loop DMD" runs that model forward from the last training frame without ever looking at a measurement. H: the rows of Phi at the sensor locations, so that H x_k is the modal prediction of the sensor readings; q_mean,s: the entries of the mean field at those same sensor locations, which must be subtracted from a raw reading before it is compared with H x_k. Kalman filter: the recursion that combines the model forecast with each new measurement, weighting them by their covariances; the Kalman gain K is that weight. Relative L2 error: ||estimate - truth|| / ||truth|| over the ROI at each test frame, averaged over frames.

Worked scalar example of one predict-update step. Suppose one mode, one sensor, F = 1, H = 1. After the previous update the estimate is x+ = 2.0 with variance P+ = 0.5, the process noise is Q = 0.5 and the sensor noise R = 1.0. Predict: x- = 2.0, P- = 0.5 + 0.5 = 1.0. A measurement y = 3.0 arrives: innovation r = 3.0 - 2.0 = 1.0, its variance S = P- + R = 2.0, gain K = P- / S = 0.5, update x+ = 2.0 + 0.5 * 1.0 = 2.5 and P+ = (1 - K) P- = 0.5. The estimate moves halfway towards the measurement because model and sensor are equally uncertain; a noisier sensor (larger R) would move it less. The matrix recursion below is this arithmetic with vectors and covariance matrices.

Why persistence scores above 100%: the transverse velocity in the wake window oscillates about zero, so a frame frozen at the last training time is out of phase with the truth for most of the test window and its error can exceed the norm of the truth itself (131.8% here). An error above 100% is therefore possible for any estimate that is anti-correlated with the truth; it is not a bookkeeping mistake.

01 Reduced model and observations

Let x_k be a **column** of POD coefficients; the code stores its transpose a_k. The POD basis contains training modes, and H selects its sensor rows. With $F = A^T$, the model is:

$$q_k = q_{mean} + \Phi x_k$$
$$x_{k+1} = Fx_k + w_k, \quad w_k \sim N(0,Q)$$
$$y_k = q_{means} + Hx_k + v_k, \quad v_k \sim N(0,R)$$

Worked orientation check: a_row = [2,3] and A = [[1,0.1],[0,1]] give a_row A = [2,3.2]. Transposing the same calculation gives F x = A^T [2,3]^T = [2,3.2]^T. Both conventions describe one transition, not two models.

A comes from training-only least squares. Q begins with the covariance of the column one-step training residuals and is multiplied by a validation-selected scale. R is fixed from the declared synthetic sensor-noise level. D-optimal

oversampling chooses sensor positions from training POD modes only.

The model is deliberately imperfect: truncation, nonlinear dynamics and biased covariance estimates remain. Covariance inflation is a model-selection parameter, not a physical noise measurement.

02 Causal predict-update recursion

$$x^- = Fx^+, \quad P^- = FP^+F^T + Q$$

$$r = y - q_{mean,s} - Hx^-, \quad S = HP^-H^T + R$$

$$K = P^-H^TS^{-1}, \quad x^+ = x^- + Kr$$

$$P^+ = (I-KH)P^-(I-KH)^T + KRK^T$$

Superscripts - and + denote predicted and updated states. The code solves for the gain and uses the Joseph covariance update. A prefix-invariance test checks that future measurements cannot alter earlier estimates.

An RTS smoother would revisit time k using observations after k. It can be valuable for retrospective denoising, but it answers a different question and would be an unfair baseline for online estimation.

Reproducibility: Required evidence: exact split, all candidates, all seeds, hashes, environment, matched baselines, common-scale contours and the retained interval failure. Run All writes only to a fresh temporary folder; tracked data/results remain unchanged.

Protocol and quantitative evidence

Frozen selection and evaluation

Training is frames 0:160; validation 160:210; test 210:281. Candidate ranks are 4, 6 and 8; sensor counts are 8, 16 and 32; process-covariance scales are 1, 10, 100 and 1000. Five validation noise seeds select minimum mean validation relative L2. Ties favor fewer sensors, lower rank and lower inflation.

Final evaluation uses five different seeds. Noise standard deviation is 10% of training-field RMS. Test truth never selects rank, sensors, inflation or noise level. The selected model is rank 8 with 32 sensors and scale 1000.

Baselines: open-loop DMD receives no measurements after frame 159; persistence repeats frame 159; sensor-only reconstruction solves a POD least-squares problem independently at every measured time. Sensor-only receives the same measurements as the filter and is therefore the strongest matched information baseline.

Test error across five measurement-noise seeds

Method	Mean relative L2 (%)	Sample SD (pp)
Kalman filter	2.179	0.066
Sensor-only POD	3.163	0.064
Open-loop DMD	4.359	0.000
Persistence	131.813	0.000

SD is measured in percentage points (pp). Open-loop and persistence ignore sensor noise, so their across-seed SD is zero. All rows use the same test frames; these are repeated measurements of one CFD trajectory.

The comparison supports a bounded claim: assimilating current noisy observations improves this retained trajectory under this declared observation model. It does not establish new-Re generalization, grid independence or field accuracy against experiment.

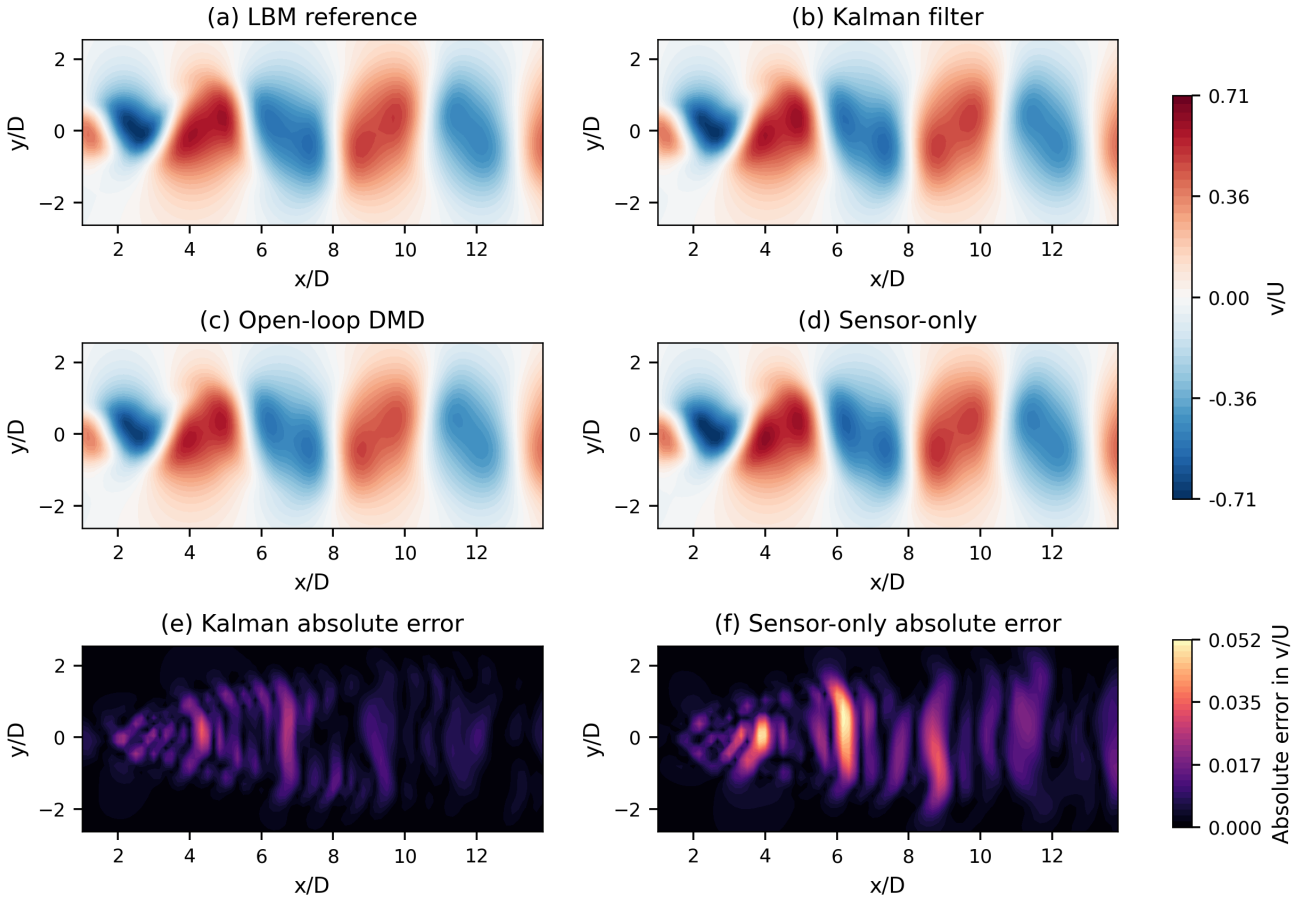
Uncertainty audit: a useful failure

Project P_k through the POD basis to obtain marginal point variances. Nominal 95% Gaussian intervals cover only 55.10% of test field values on average. Mean full interval width is 0.00760 v/U. This is severe under-coverage, even though the point estimate is accurate.

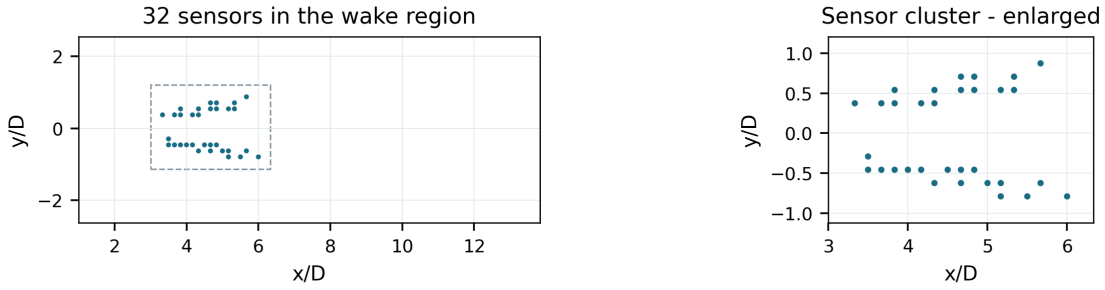
Why? Training residual covariance omits structural model error; modes truncate the state; Gaussian white-noise assumptions are incomplete; pointwise field errors are correlated. Validation selected point error, not coverage. We retain the failure rather than calibrating on test data.

Marginal pointwise coverage is descriptive, not simultaneous field coverage. A next study may choose inflation or conformal calibration on validation coverage, then freeze it before a genuinely untouched trajectory.

Wake reconstruction and local errors



Final test frame, first evaluation seed (31). Panels (a-d) share the velocity scale; (e-f) share the absolute-error scale. Equal x/D and y/D scales preserve physical geometry. Scores use the original 32 x 78 samples. The cylinder lies outside the plotted region.



Sensor locations are shown separately. The dashed rectangle identifies the enlarged view; every dot denotes one of the same 32 training-selected locations.

Exercises and references

Exercises: derive the row/column covariance convention; explain why sensor-only can beat a poor dynamical model; show why a smoother is non-causal; design a missing-sensor stress test; predeclare a new-Re evaluation and coverage criterion.

Canonical reference: S. Sarkka and L. Svensson, Bayesian Filtering and Smoothing, 2nd ed., Cambridge University Press, 2023, doi:10.1017/9781108917407.

Data: FlowMLLab cylinder-cfd-v1. Code is independently authored for this repository. No restricted handout text, code, figure, exercise or solution is incorporated.